

23-S2-Q2

Q: class 1 & -1

(a) $x(N)$

$d(N)$

SVM formulate the primal problem
separable

(2) SVM $\rightarrow x$.

$$S_1 = [0.2575 \quad 0.0598 \quad 0.7212 \quad 0.8555]^T$$

$$S_2 = [4.5052 \quad -0.4733 \quad 3.3539 \quad 4.6830]^T$$

$$S_3 = [3.5651 \quad -1.3827 \quad 2.8324 \quad 2.6521]^T$$

$$S_4 = [-0.2015 \quad -2.1464 \quad 1.0908 \quad 1.5509]^T$$

Solution (a) Given the training sample $\{x(i), d(i)\}_{i=1,2,\dots,N}$ find the optimal values of the weight vector w and bias b such that they satisfy the constraints

$$d(i) \times [w^T x(i) + b] \geq 1 \quad \text{for } i = 1, 2, \dots, N$$

and the weight vector minimizes the following cost function

$$J(w) = \frac{1}{2} w^T w$$

(b) ① calculate w^*

$$\begin{aligned} w^* &= \sum_{i=1} \alpha(i) d(i) s(i) \\ &= 0.3646 \times 1 \times [0.2571 \quad 0.0598 \quad 0.7212 \quad 0.8555]^T \\ &\quad + 0.0067 \times (-1) \times [4.5052 \quad -0.4733 \quad 3.3139 \quad 4.6830]^T \\ &\quad + 0.029 \times (-1) \times [3.5651 \quad -1.3827 \quad 2.8324 \quad 2.6521]^T \\ &\quad + 0.3289 \times (-1) \times [-0.2015 \quad -2.1464 \quad 1.0908 \quad 1.5509]^T \\ &= [0.02659 \quad 0.7710 \quad -0.2004 \quad -0.3065]^T \end{aligned}$$

② calculate b^*

$$\begin{aligned} b^* &= \frac{1}{N_s} \sum_{\alpha(i) \neq 0} [d(i) - (w^*)^T x(i)] \\ &= \frac{1}{4} \times [1 - (w^*)^T x(1) - 1 - (w^*)^T x(2) - 1 - (w^*)^T x(3) - 1 - (w^*)^T x(4)] \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{4} \left\{ -2 - (w^*)^T [x(1) + x(2) + x(3) + x(4)] \right\} \\
&= \frac{1}{4} \left\{ -2 - [0.02659 \quad 0.7710 \quad -0.2004 \quad -0.3065] \begin{bmatrix} 8.1263 \\ -3.9426 \\ 7.9983 \\ 9.7415 \end{bmatrix} \right\} \\
&= 1.3531
\end{aligned}$$

$$\textcircled{3} y = w^T x + w_0$$

$$= (w^*)^T x + b^*$$

$$\begin{aligned}
&= [0.02659 \quad 0.7710 \quad -0.2004 \quad -0.3065] \begin{bmatrix} -0.2031 \\ -2.2553 \\ 2.6182 \\ 2.6170 \end{bmatrix} + 1.3531 \\
&= -1.7179 < 0
\end{aligned}$$

predicted class = -1

(c) Fisher's Linear Discriminant Analysis (LDA)

Linear SVM

① criterion

LDA: find such w that the following criterion is maximized

$$J(w) = \frac{(\tilde{m}_1 - \tilde{m}_2)^2}{\tilde{S}_1^2 + \tilde{S}_2^2}$$

SVM: find the weight vector minimizes the following cost function $J(w) = \frac{1}{2} w^T w$

② philosophy

LDA: max the ratio $\frac{w^T S_B w}{w^T S_W w}$

SVM: max the margin between the closest points of opposite classes

③ Data Usage

LDA: all sample mean and covariances influence w

SVM: Only support vectors determine w

④ Optimisation

LDA: Closed-form eigen vector solution for two classes

SVM: convex quadratic programme solved via Lagrange

⑤ Summary

LDA: optimises class separation in a probabilistic sense and Gaussian assumption

SVM: optimises geometric margin with minimal distributional assumption.